

Josephson anomalous vortices

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We show that vortices with circulating current, related with odd-frequency triplet pairing, appear in Josephson junctions where the barrier is a weak ferromagnet with strong spin-orbit coupling. By symmetry analysis we show that there is an additional term—a rotary invariant—in the superconducting free energy which allows for magnetoelectric effects even when the previously considered Lifshitz invariant vanishes. Using a microscopic model based on a modified Usadel equation incorporating those effects, we show that the size, shape, and position of these vortices can be controlled by manipulating Rashba spin-orbit coupling in the weak link, via gates, and we suggest that these vortices could be detected via scanning magnetometry techniques. We also show that the transverse triplet components of the pairing amplitudes can form a texture.

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Introduction. Vortices in superconductors have remained a vibrant area of research since their discovery in the 1950s. In the original work by Abrikosov [1], vortices appear in type II superconductors threaded by a flux quantum as a result of the superconducting phase winding by 2π around a singularity. This produces a signature circulating current with a normal core. Superconducting heterostructures allow for a rich variety of other kinds of vortices. For example, in Josephson junctions coreless, elongated vortices can arise due to solitons in the superconducting phase [2]. Vortices with cores can also appear in normal metals in proximity to a superconductor [3], and also in junctions due to destructive interference of the pairing amplitudes [4]. Swirling supercurrents and vortices can appear in structures with strong spin-orbit coupling (SOC), for example when there are magnetic impurities [5] or when an in-plane exchange field is applied [6,7]. In these cases, spin-triplet pairing amplitudes are created by the exchange coupling, and modulated by spin-orbit coupling. In analogy to superfluid He-3 [8–10], it might be possible to also generate more exotic vortex matter in superconducting systems with such triplet pairing amplitudes.

Here we introduce the Josephson anomalous vortex (JAV). These appear as a circulating current density [Fig. 1(b)] in two-dimensional junctions where the Josephson weak link is ferromagnetic and has strong SOC [Fig. 1(a)], as long as no current is applied. JAVs only appear when the weak link, or gate-defined regions in the weak link, are close to a coherence length in length and width. The current density is at a maximum at the edges of the vortex, and vanishes at the center. Correspondingly, at the center of a vortex the triplet part of the pairing amplitudes vanishes, leaving only the singlet

part; these vortices do not have a core. These amplitudes are textured in a manner similar to a magnetic vortex.

In this Letter we study a rotary invariant in the free energy. The rotary invariant is related to a boundary term, and so in the absence of an applied current generates circulating currents. Circulating currents due to the rotary invariant arise even when the standard Lifshitz invariant [11] vanishes. We show that JAVs are not fine tuned so that they can appear over a range of spin-orbit and exchange energies. We argue that, due to the magnetic field produced by swirling currents, JAVs could be detected by scanning magnetometry techniques. We also suggest that these vortices may be manipulated by Rashba “islands” within the weak link via back gates.

Throughout this Letter we use the singlet superconducting gap Δ_0 of the proximity induced superconductivity in the ferromagnet as the natural unit of energy, and the superconducting coherence length ξ_0 as the unit of length. We restore the physical units when predicting amplitudes of measurable quantities. In numerical calculations we fix the temperature to $T = 0.1 \Delta_0 / k_B$. We use the conventions that Greek subscripts

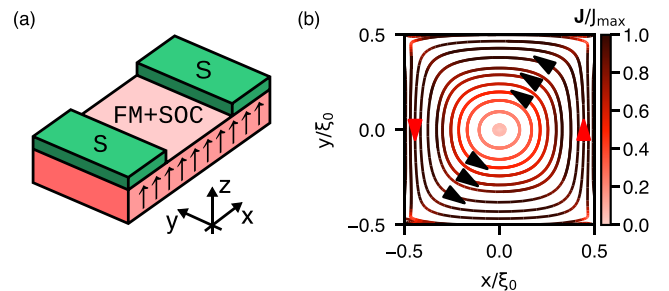


FIG. 1. (a) Sketch of a superconductor-ferromagnetic metal-superconductor device studied here. There is strong spin-orbit coupling in the ferromagnetic layer. (b) Circulating current density in the weak link of (a) due to the rotary invariant, computed numerically using the microscopic model. Here no current is applied across the junction. Parameters: $(L_x, L_y, J, \alpha) = (\xi_0, \xi_0, \Delta_0, 2.35/\xi_0)$.

μ, ν run over $(0, 1, 2, 3)$, while Latin subscripts run over $(1, 2, 3)$. Repeated indices imply summation.

Magnetoelectric effects without a Lifshitz invariant. Over the past decades, it has been shown experimentally and theoretically that SOC allows for coupling between the magnetic and electric properties of a material [11]. Such coupling is expressed in phenomena such as the superconducting and Josephson diode effects [12], helical phases [13], and the Edelstein and inverse Edelstein effects [14–16]. For weak superconductivity, many magnetoelectric effects can be described by the addition of a Lifshitz invariant in the Ginzburg-Landau free energy [11]. In this Letter we show first from symmetry that magnetoelectric effects can give rise to boundary invariants in addition to the bulk Lifshitz invariant in the most generic free energy. The most interesting is an invariant that we term the rotary invariant (reported earlier in Ref. [17] but with its significance not discussed). Such rotary invariants can appear in a wide range of superconducting systems when there is no Lifshitz invariant, as long as the relevant symmetries are satisfied, and generally they describe vortex like excitations appearing in inhomogeneous systems or systems with boundaries. With insight from this general symmetry analysis, we apply a concrete microscopic model (using quasiclassical methods [18]) describing disordered superconductors in the presence of spin-orbit coupling, derive the microscopic expressions for the invariants, and study the vortices in parameter regimes much beyond the Ginzburg-Landau theory.

Consider first the superconducting free energy density:

$$\mathcal{F} = \mathcal{F}_S + \mathcal{F}_L + \mathcal{F}_R. \quad (1)$$

Here $\mathcal{F}_S$ is the usual Ginzburg-Landau free energy [19],

$$\mathcal{F}_S = a|\psi|^2 + \frac{b}{2}|\psi|^4 + \frac{1}{2}D_{jk}\Pi_j\psi(\Pi_k\psi)^* + \frac{|\mathbf{B}|^2}{2\mu_0}, \quad (2)$$

with ψ the complex order parameter of the superconductor, a and b the Ginzburg-Landau constants, D_{jk} the superfluid stiffness symmetric tensor, and $\mathbf{B}$ the magnetic field. For compactness we define the momentum $\Pi_j = -i\hbar\hat{\nabla}_j = -i\hbar\partial_j - 2eA_j$, with A_j the electromagnetic vector potential. If time-reversal symmetry is broken, and inversion symmetry intrinsically broken, we can add d_k , the Lifshitz invariant vector, via the second term $\mathcal{F}$ [20], with

$$\mathcal{F}_L = d_k[\psi^*\Pi_k\psi + \psi(\Pi_k\psi)^*]. \quad (3)$$

If time-reversal symmetry is broken and the inversion symmetry is broken by extrinsic inhomogeneity (e.g., via system boundaries, see the Appendix) we can add the rotary invariant e_{jk} , an antisymmetric tensor, via the third term $\mathcal{F}_R$, with

$$\mathcal{F}_R = ie_{jk}\Pi_j\psi(\Pi_k\psi)^*. \quad (4)$$

One mechanism for breaking time-reversal and inversion symmetry comes from the combination of SOC and exchange field $\mathbf{h}$, which we consider here. In the case of linear in momentum SOC, with $H_{SO} = A_j^a k_j \sigma^a / (2m)$, we can define the SU(2) fields $\mathcal{A}_j = A_j^a \sigma^a$ [21,22] and the corresponding field strength tensor $F_{ij} = -i[\mathcal{A}_i, \mathcal{A}_j]$, valid for position independent SU(2) fields. Using these definitions, we find that (see Sec. II in the

Supplemental Material [23]) generally

$$d_k = i\kappa \text{Tr}\{F_{kj}[\mathcal{A}_j, h_i\sigma_i]\}, \quad e_{jk} = \kappa \text{Tr}[F_{jk}h_i\sigma_i], \quad (5)$$

where $\kappa = -\frac{7\zeta(3)v_F\eta}{16\pi^2k_B^2T_c^2}$, with v_F the density of states per spin projection at the Fermi level, $\eta = D\ell^2/(k_F\ell)$, $D = v_F\ell/2$ the diffusion constant, v_F the Fermi velocity, k_F the Fermi wave vector, T_c is the critical temperature, and ℓ the mean free path. $\zeta(n)$ is the Zeta function. Thus d_k is cubic and e_{jk} quadratic in the strength of SOC (here we assume weak SOC; in contrast, when $\alpha k_F \gg \Delta_0$ the Lifshitz invariant d_k is linear in SOC strength α [13]). Moreover, these forms imply that d_k is nonzero only when the exchange field and F_{ij} have perpendicular components, whereas e_{jk} requires the presence of their parallel components. We can hence study the implications of one without the other.

To illustrate this point, we focus on the junction geometry shown in Fig. 1(a). Here we have an exchange field $\mathbf{h} = J\hat{z}$, and Rashba SOC due to an electric field also in the z direction. The SOC Hamiltonian is $H_{SO} = \alpha(\hat{z} \times \mathbf{k}) \cdot \boldsymbol{\sigma}$ and the corresponding field-strength tensor is $F_{xy} = -F_{yx} = 2\alpha^2\sigma_z$. Hence, the Lifshitz invariant vanishes $d_k = 0$, and the rotary invariant is finite $e_{xy} = -e_{yx} = 4\kappa\alpha^2J$. Integration by parts (see the Appendix) of this rotary term results in a boundary term $\propto n_j e_{jk}[\psi\partial_k\psi^* - \text{c.c.}]$ that generates circulating currents perpendicular to the boundary normal n_j .

To investigate the rotary invariant outside of the Ginzburg-Landau regime, for the rest of this Letter we study a microscopic model for the heterostructure sketched in Fig. 1(a), using the methods described in Refs. [18,24,25] (see Sec. I in the Supplemental Material [23] for details). From the preceding analysis we expect a finite rotary invariant (and the related magnetoelectric effects) for this geometry because the relevant symmetries are satisfied. We assume that the ferromagnet is weak, so that the quasiclassical approximation is valid (i.e., the density of states is the same for both up and down spins). Note that we assume that the superconducting electrodes proximity induce superconductivity into the ferromagnetic thin film region just below the electrodes. In general this gap Δ_0 is smaller than the zero-temperature, zero-field gap of the electrodes. Spin-orbit coupling also increases the critical field, so we can study the effects of large exchange field, $J \sim \Delta_0$ [26].

Due to the exchange field, the order parameter (i.e., pairing amplitude or anomalous Green's function) for this system is not scalar, but rather has a spin structure, $\mathbf{f} = f_s\sigma_0 + f_i\sigma_i$ [27]. Here f_s is the singlet part and $f_i = \mathbf{f}_i = (f_x, f_y, f_z)$ the triplet part of the anomalous Green's function. As expected from the Ginzburg-Landau analysis, this microscopic model also has a rotary invariant. For this model, the rotary invariant involves the f_s and f_z components of the pairing amplitude (see Sec. I in the Supplemental Material [23]),

$$\mathcal{F}_{\text{rot}} = i\zeta(\hat{\nabla}f_s \times \hat{\nabla}^*f_z^*) \cdot \hat{z}. \quad (6)$$

Here $\zeta = \pi v_F \hbar \eta \alpha^2$; in all numerical results we assume $\hbar\eta = 0.01\Delta_0\xi_0^4$. Note that this term scales as $\sim 1/E_F$ and so breaks the quasiclassical symmetry [24,28,29] (that is, the combined time-reversal and electron-hole symmetries). The rotary term $\mathcal{F}_{\text{rot}}$ appears as long as there is both SOC and an exchange field, even when there is no Lifshitz invariant. It generates

circulating current densities,

$$\mathbf{J}_{\text{rot}} = 2e\zeta(f_s \text{curl} f_z^* + f_s^* \text{curl} f_z), \quad (7)$$

with $\text{curl} A = (\partial_y A, -\partial_x A)$ and e the elementary charge. Here f_s varies across the junction due to the proximity effect and f_t is in general finite due to the presence of the exchange field. We expect the conventional Ginzburg-Landau terms to dominate over the rotary invariant, so to see the effects of this term there should be no current (i.e., phase bias φ) across the junction.

We emphasize that these circulating currents are distinct to those introduced in Ref. [30], which are due to an interfacial SOC generated surface Lifshitz invariant. While these surface currents can be similar, we consider only the bulk effect here. This depends on the geometry: in Fig. 1(a), the FM/S interface normal is parallel to the exchange field, and the interfacial Lifshitz invariant vanishes. In contrast, in a lateral junction (where the ferromagnetic weak link is sandwiched between two superconductors on the left and right), both the interface and the bulk effects would matter. We additionally show in Sec. III of the Supplemental Material [23] that vortices can form on a superconducting disk due to the rotary invariant.

In Fig. 1(b) we plot the current density for a junction of $L_x \times L_y = (1 \times 1)\xi_0^2$, at zero phase bias. This yields a representative current vortex, distinguished by the strong current density at the boundary (as opposed to the center, as in an Abrikosov vortex) and vanishing current in the center. We compute numerically current densities and pairing amplitudes for the microscopic model using the code described in Ref. [17]. For large junctions we find some swirling current close to the superconducting terminals (see Sec. IV in the Supplemental Material [23], and Ref. [31] therein), similar to what has been reported in Ref. [7] for in-plane exchange fields. As long as $\alpha^2 \xi_0^2 \gg \varphi$, the rotary invariant, and thus the circulating current, dominate over the usual Josephson current. For $\varphi \gtrsim \alpha^2 \xi_0^2$ the rotary invariant perturbs the Josephson current (see Sec. V in the Supplemental Material [23]).

Antivortex patterns and superconducting textures. Often current vortices in superconductors are accompanied by textures in the phase of the superconducting order parameter; in this case, the anomalous Green's function (pairing amplitude) f . Here for all finite exchange fields J and Rashba spin-orbit coupling strengths α tested, we find numerically that the triplet part of the pairing amplitudes is textured. In Fig. 2(a) we plot f_x, f_y as a vector field, forming a swirling pattern reminiscent of a magnetic texture. Equivalently, we plot in Fig. 2(b) the complex phase of the $f_x + if_y$ anomalous Green's functions, featuring a phase winding reminiscent of an Abrikosov vortex. Finally, we plot in Fig. 2(c) the phase texture of the f_z triplet. This texture does not completely wind, due to the line defect along $x = 0$. All three textures presented are representative, with only small variations with J, α . In all our simulations, the amplitudes scale like $|f_s| > |f_z| > |f_x| \sim |f_y|$. From Eq. (7) it is clear that the f_z texture generates current vortex patterns [Fig. 2(f)]. In addition to vortices we also find antivortex patterns [Fig. 2(d)] [32] in the current density, and mixtures between the two [Fig. 2(e)]. As with the vortices, the current vanishes in the center of an antivortex pattern. The appearance of the latter patterns is due

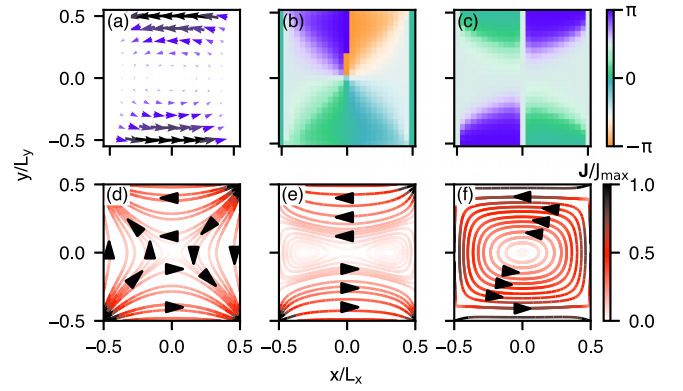


FIG. 2. (a) Texture formed from the anomalous Green's functions f_x, f_y , reminiscent of a magnetic texture. (b) Phase winding of the anomalous Green's functions $f_x + if_y$, reminiscent of an Abrikosov vortex. (c) Phase texture of the superconducting correlation f_z . (d) Representative antivortex pattern in the current density. (e) Representative mixture between a current antivortex and a vortex pattern. (f) Representative vortex. Parameters for all panels: $(L_x, L_y, J) = (\xi_0, \xi_0, \Delta_0)$; for (a)–(c), (f): $\alpha = 2/\xi_0$; for (d): $\alpha = 1/\xi_0$; for (e): $\alpha = 1.55/\xi_0$.

to the influence of the f_x, f_y textures, and depends subtly on the small variations of f_t with J, α .

Because the components of the pairing amplitude respect the 180° rotation symmetry of the junction, the triplets f_t change sign across the origin, and so the triplets (and thus the current) vanish at the origin. However, the singlet f_s does not change sign when no current is applied due to the symmetric boundary conditions. Thus, the singlet is finite at the origin and these vortices are coreless.

The most natural way to categorize singularities in vector fields is with the Poincaré index. This is the winding number [33,34]

$$\mathcal{P} = \frac{1}{\pi} \oint_C d\theta, \quad (8)$$

with θ the angle a vector field (such as the current density) relative to the x axis, and C a loop around a singularity. When there are multiple singularities, the total index is the sum of all $\mathcal{P}$ for each singularity. Note that because the current density does not vanish far from the singularity, we see a continuous transition between antivortex and vortex patterns, rather than a first-order transition.

To gain some intuition for this index, consider the current densities $\mathbf{J}_V$ and $\mathbf{J}_{AV}$, which approximate the vortex and antivortex patterns (at least not too close to the boundaries),

$$\mathbf{J}_V = J_0 \left(\frac{-y}{\xi_0} \hat{x} + \frac{x}{\xi_0} \hat{y} \right), \quad \mathbf{J}_{AV} = J_0 \left(\frac{y}{\xi_0} \hat{x} + \frac{x}{\xi_0} \hat{y} \right). \quad (9)$$

Here $J_0 = e\zeta \Delta_0 / (\hbar \xi_0)$. By choosing the integration path C to be a circle, a straightforward calculation shows that $\mathcal{P} = -1$ for antivortex and $\mathcal{P} = +1$ for vortex patterns—this is true regardless of the vorticity.

We plot in Figs. 3(a)–3(c) the total index (computed from the numerically obtained current densities) over a range of J, α , for three different system sizes, $L_x \times L_y = \{0.5 \times 0.5, 1 \times 1, 5 \times 5\} \xi_0^2$. For the systems smaller than

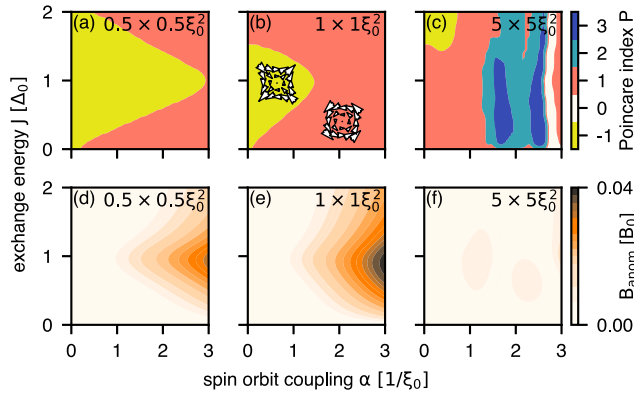


FIG. 3. (a)–(c) Total Poincaré index $\mathcal{P}$. Antivortex patterns in the current correspond to $\mathcal{P} = -1$ and vortices to $\mathcal{P} = +1$, as indicated by the inset sketches. Junction size is indicated in the top right hand corner, $L_x \times L_y \xi_0^2$. (d)–(f) Magnetic field strength along the $\hat{z}$ direction at the center of the junction, for the same parameter spaces.

$L_x \times L_y = (5 \times 5) \xi_0^2$ there are only two phases, vortex and antivortex (as sketched in the inset). For the largest system tested, $(5 \times 5) \xi_0^2$, the phase diagram is richer. For $\alpha > 1.25/\xi_0$ the total index $\mathcal{P}_{\text{tot}} > 1$, due to the presence of multiple singularities [23]. For $\alpha < 1.25/\xi_0$ there are single vortices and antivortices in the current density.

Using the Biot-Savart law, we plot in Figs. 3(d)–3(f) the magnetic field at the center of the vortex (also from the numerics), for the same parameter ranges as (a)–(c). The magnetic field scale in the plot is $B_0 = \mu_0 \sigma_D \Delta_0 t / (4e \xi_0) = 1 \text{ mT} \times (t/\xi_0) [\sigma_D / (1.6 \times 10^7 \text{ S/m})] [\Delta_0 / (200 \text{ } \mu\text{eV})]$, expressed in terms of relatively typical values for the Drude conductivity $\sigma_D = v_F e^2 D$ in disordered metals, and a typical Δ_0 for Al. We also include the thickness $t \ll \xi_0$ of the junction. We do not compute this magnetic field self-consistently (i.e., by feeding back into the electromagnetic vector potential). As a comparison, we can apply the Biot-Savart law to the approximation Eq. (9), yielding the magnetic field at the center of the vortex

$$B_{\text{anom}} = \mu_0 J_0 \frac{Rt}{2\xi_0}, \quad (10)$$

with μ_0 the vacuum permeability, and R the radius of the vortex. Substituting in $R = \xi_0$, we find a maximum magnetic field strength of $B_{\text{anom}} = 2\pi(R/\xi_0)(\alpha\xi_0)^2(\eta/\Delta_0\xi_0^4)B_0$, some 1.5 times larger than what is seen in numerics for vortices in the $(1 \times 1) \xi_0^2$ system.

For the $(1 \times 1) \xi_0^2$ system, which hosts vortices for most of parameter space, we see the largest B_{anom} , increasing with increasing α , and peaking at $J \sim \Delta_0$. This magnetic field is due to the vortices. For the $(0.5 \times 0.5) \xi_0^2$ system we see a weaker magnetic field, also increasing with α and peaked at $J \sim \Delta_0$. This magnetic field is due to the mixture of vortex and antivortex patterns around the phase transition. For the $(5 \times 5) \xi_0^2$ system there is almost no magnetic field. In general, most of the current density is concentrated close to the superconducting leads, so in the vortex phase $\alpha < 1.25/\xi_0$ there is very little current and hence only a tiny magnetic field generated. When there are multiple singularities, $\alpha > 1.25/\xi_0$, the currents around each singularity counterpropagate, resulting in the overall magnetic field vanishing (see

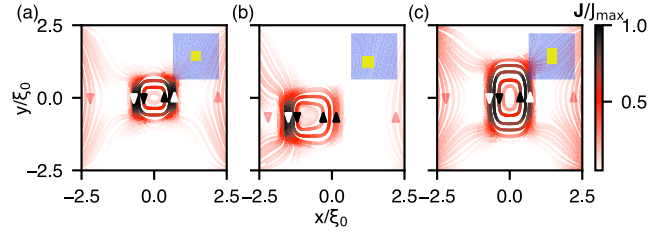


FIG. 4. The size (a), position (b), and shape (c) of a JAV could be manipulated by defining an island with strong Rashba SOC α_{island} via gates, within a weak link with weak SOC α_{back} . Inset: sketches of the gate defining α_{island} (yellow patch) within the weak link (blue square). Parameters: $(L_x, L_y, J, \alpha_{\text{island}}, \alpha_{\text{back}}) = (5\xi_0, 5\xi_0, \Delta_0, 2/\xi_0, 1/(2\xi_0))$.

Sec. IV in the Supplemental Material [23]). For (e)–(g), the peak in B_{anom} around $J = \Delta_0$ is due to the fact that $f_z \sim J/(J^2 + \omega^2)$ (with ω the Matsubara frequency) [23]—i.e., the magnetic field must peak at some finite J and then decay for large J .

Characterization and manipulation. Because circulating currents produce a uniform magnetic field, a natural approach to characterizing these vortices is via nanomagnetometry. Scanning probe techniques are well established and can operate at cryogenic temperatures with high spatial resolution. Magnetic force microscopy has already been used to study conventional Josephson vortices [35,36], and can operate with a resolution of 10–50 nm [37], or even atomic resolution [38]. Scanning nitrogen-vacancy center [39] and nanoSQUID methods [40] can now both achieve spatial resolutions of around 50 nm; indeed, recently, scanning nitrogen-vacancy centers were used to study vortices in NbSe₂ [41]. These magnetometry techniques are complicated by the need to subtract the background ferromagnetic signal. One possible solution is to include a back gate to control the Rashba interaction [42]. One could then effectively switch off SOC to isolate the background ferromagnetic signal. The absence of SOC can, for example, be confirmed via the absence of a Josephson diode effect.

Such gate-based techniques for controlling Rashba SOC could also be used to manipulate these vortices. Consider a large weak link which has only weak (or absent) Rashba SOC. One can define an island within the weak link with strong Rashba SOC via a small back gate. We show in Fig. 4 examples where such Rashba “islands” can be used to control the size, position, and shape of a JAV. Such methods could be used to realize JAVs in arbitrarily sized junctions, overcoming the necessity of defining a square geometry of a coherence length. Indeed, several different vortices could be defined within a single weak link using this technique.

Outlook. We describe the characterization and manipulation of Josephson anomalous vortices, which appear in Josephson junctions where the weak link is a weak ferromagnet with strong spin-orbit coupling. The exchange field and spin-orbit coupling conspire to lead to circulating currents that we associate phenomenologically to a previously unexplored rotary term in the free energy. The same ingredients lead to textures in the triplet components $\mathbf{f}_t$.

Josephson anomalous vortices could be found in weak links based on heterostructures such as ferromagnetic thin

film/normal metal bilayers, ferromagnetic alloys deposited as thin films on metals, or two-dimensional van der Waals heterostructures, coupled to conventional superconducting electrodes. Possible thin film bilayers include YIG/InAs or EuS/InAs [43], which must be tuned so that the magnetization turns out of plane (see, e.g., Ref. [44] and references therein). Candidate ferromagnetic alloys include Pd/Ni [45] or Cu/Ni alloys [46]. The layers must be chosen carefully to minimize the Dzaloshinskii-Moriya interaction, which will produce a magnetic texture rather than a pure out-of-plane exchange field. Finally, we suggest that the van der Waals monolayer magnet CrI₃, which has out-of-plane magnetization [47], could be combined with other van der Waals materials with strong Rashba spin-orbit coupling, such as MoS₂ or WSe₂ [48], to find the anomalous vortices.

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Data availability. The data that support the findings of this article are openly available [49].

Appendix: Lifshitz and rotary invariants in Ginzburg-Landau theory. Here we derive the rotary invariant in the Ginzburg-Landau picture phenomenologically, and discuss its meaning in more detail.

The expression for the free energy must remain unchanged with respect to various global transformations that leave the physical system invariant [20]. The transformations of the order parameter field are $\psi \mapsto \psi^*$ (time reversal $\mathcal{T}$, TR), $\psi(\mathbf{r}) \mapsto \psi(-\mathbf{r})$ (inversion $\mathcal{I}$), $\psi(\mathbf{r}) \mapsto \psi(R_{ij}r_j)$ (global rotation), and $\psi(\mathbf{r}) \mapsto e^{i\varphi}\psi(\mathbf{r})$ [global $U(1)$]; coefficients transform keeping $\mathcal{F}$ invariant and real. Further restrictions would come from crystal symmetry, but we do not consider that here.

Expanding in gradients and expressing the result in terms of real coefficients, the possible terms with one gradient are

$$\mathcal{F}_1 = d_k i[\psi \partial_k \psi^* - \text{c.c.}] + c_k \partial_k |\psi|^2. \quad (\text{A1})$$

Then, d_k is a TR-odd vector ($\mathcal{T}, \mathcal{I} = -, -$) and c_k a TR-even vector ($\mathcal{T}, \mathcal{I} = +, -$). From integration by parts, the c_k term is equivalent to the Ginzburg-Landau $a|\psi|^2$ term, and a similar boundary term $n_k c_k |\psi|^2$ where n_k is the boundary normal vector.

The possible terms with two gradients are

$$\mathcal{F}_2 = D_{jk} \partial_j \psi \partial_k \psi^* + i e_{jk} \partial_j \psi \partial_k \psi^* + f_{jk} \partial_j \partial_k |\psi|^2 + g_{jk} i \partial_j [\psi \partial_k \psi^* - \text{c.c.}], \quad (\text{A2})$$

where without loss of generality we have taken $D_{jk} = D_{kj}$, $e_{jk} = -e_{kj}$, and $f_{jk} = f_{kj}$, $g_{jk} = g_{kj}$. Then, D_{jk} is a TR-even tensor ($\mathcal{T}, \mathcal{I} = +, +$) and e_{jk} a TR-odd tensor ($\mathcal{T}, \mathcal{I} = -, +$). Similarly, f_{jk} is a TR-even and g_{jk} a TR-odd tensor.

For spatially constant coefficients, the terms aside from the superfluid stiffness D_{jk} part are total derivatives. From integration by parts, the f_{jk} term is equivalent to the c_k term and a boundary term $n_j f_{jk} \partial_k |\psi|^2$. The other part g_{jk} is equivalent to a Lifshitz invariant $(\partial_j g_{jk}) i[\psi \partial_k \psi^* - \text{c.c.}]$ and a similar boundary term $n_j g_{jk} i[\psi \partial_k \psi^* - \text{c.c.}]$. The contribution from the e_{jk} term also reduces as

$$\int_M i e_{jk} \partial_j \psi \partial_k \psi^* = - \int_{\partial M} \frac{1}{2} n_j e_{jk} i[\psi \partial_k \psi^* - \psi^* \partial_k \psi] \quad (\text{A3})$$

$$+ \int_M \frac{1}{2} (\partial_j e_{jk}) i[\psi \partial_k \psi^* - \psi^* \partial_k \psi]. \quad (\text{A4})$$

Consequently, e_{jk} is also equivalent to a bulk Lifshitz invariant and a similar boundary term.

On a phenomenological level, the e_{jk} term can then always also be expressed in terms of a boundary term. However, as shown in Sec. I of the Supplemental Material [23], an equivalent term arises naturally as a bulk contribution in the microscopic theory, which is valid for arbitrary smooth spatially varying gauge and exchange fields (i.e., on scales longer than the mean free path). For interfaces described by such theory, the effective boundary Lifshitz invariant that is responsible for the vorticity is microscopically generated by the bulk e_{jk} term.

- [1] A. A. Abrikosov, The magnetic properties of superconducting alloys, *J. Phys. Chem. Solids* **2**, 199 (1957).
- [2] A. V. Ustinov, Solitons in Josephson junctions, *Phys. D* **123**, 315 (1998).
- [3] V. S. Stolyarov, T. Cren, C. Brun, I. A. Golovchanskiy, O. V. Skryabina, D. I. Kasatonov, M. M. Khapaev, M. Y. Kupriyanov, A. A. Golubov, and D. Roditchev, Expansion of a superconducting vortex core into a diffusive metal, *Nat. Commun.* **9**, 2277 (2018).
- [4] D. Roditchev, C. Brun, L. Serrier-Garcia, J. C. Cuevas, V. H. L. Bessa, M. V. Milošević, F. Debontridder, V. Stolyarov, and T. Cren, Direct observation of Josephson vortex cores, *Nat. Phys.* **11**, 332 (2015).
- [5] S. S. Pershoguba, K. Björnson, A. M. Black-Schaffer, and A. V. Balatsky, Currents induced by magnetic impurities in supercon-

- ductors with spin-orbit coupling, *Phys. Rev. Lett.* **115**, 116602 (2015).
- [6] M. Amundsen and J. Linder, Supercurrent vortex pinball via a triplet cooper pair inverse Edelstein effect, *Phys. Rev. B* **96**, 064508 (2017).
- [7] F. S. Bergeret and I. V. Tokatly, Theory of the magnetic response in finite two-dimensional superconductors, *Phys. Rev. B* **102**, 060506(R) (2020).
- [8] R. Rantanen and V. B. Eltsov, Transitions in vortex skyrmion structures in superfluid ³He-A driven by an analog of the zero-charge effect, *Phys. Rev. B* **107**, 104505 (2023).
- [9] R. Rantanen and V. Eltsov, Competition of vortex core structures in superfluid ³He-B, *Phys. Rev. Res.* **6**, 043112 (2024).
- [10] R. Rantanen, E. Thuneberg, and V. Eltsov, Structure of a single-quantum vortex in ³He-A, *J. Low Temp. Phys.* **220**, 88 (2025).

- [11] D. F. Agterberg, Magnetoelectric effects, helical phases, and FFLO phases, in *Non-Centrosymmetric Superconductors: Introduction and Overview*, edited by E. Bauer and M. Sigrist (Springer, Berlin, Heidelberg, 2012), pp. 155–170.
- [12] M. Nadeem, M. S. Fuhrer, and X. Wang, The superconducting diode effect, *Nat. Rev. Phys.* **5**, 558 (2023).
- [13] J. Hasan, D. Shaffer, M. Khodas, and A. Levchenko, Supercurrent diode effect in helical superconductors, *Phys. Rev. B* **110**, 024508 (2024).
- [14] V. M. Edelstein, Magnetoelectric effect in polar superconductors, *Phys. Rev. Lett.* **75**, 2004 (1995).
- [15] V. M. Edelstein, Magnetoelectric effect in dirty superconductors with broken mirror symmetry, *Phys. Rev. B* **72**, 172501 (2005).
- [16] O. Dimitrova and M. V. Feigel'man, Theory of a two-dimensional superconductor with broken inversion symmetry, *Phys. Rev. B* **76**, 014522 (2007).
- [17] P. Virtanen, Magnetoelectric effects in diffusive two-dimensional superconductors studied by the nonlinear σ model, *Phys. Rev. B* **111**, 024510 (2025).
- [18] P. Virtanen, F. S. Bergeret, and I. V. Tokatly, Magnetoelectric effects in superconductors due to spin-orbit scattering: Nonlinear σ -model description, *Phys. Rev. B* **104**, 064515 (2021).
- [19] M. Tinkham, Ginzburg-Landau theory, in *Introduction to Superconductivity* (Dover, New York, 2004), pp. 110–146.
- [20] L. D. Landau and E. M. Lifshitz, Phase transitions of the second kind, in *Course of Theoretical Physics, Statistical Physics* (Pergamon Press Ltd, Oxford, 1958), Vol. 5, pp. 424–454.
- [21] C. Gorini, R. Raimondi, and P. Schwab, Onsager relations in a two-dimensional electron gas with spin-orbit coupling, *Phys. Rev. Lett.* **109**, 246604 (2012).
- [22] F. S. Bergeret and I. V. Tokatly, Spin-orbit coupling as a source of long-range triplet proximity effect in superconductor-ferromagnet hybrid structures, *Phys. Rev. B* **89**, 134517 (2014).
- [23] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/nsdg-sn1b> for derivation of the rotary invariant from the microscopic nonlinear σ model; analytic derivation of the prefactors for the Lifshitz and rotary invariants; a brief discussion on the formation of a vortex due to the rotary invariant on a superconducting disk; additional plots highlighting the transition between vortices and antivortices as a function of α and system size; and additional plots showing the effect of finite phase bias.
- [24] P. Virtanen, F. S. Bergeret, and I. V. Tokatly, Nonlinear σ model for disordered systems with intrinsic spin-orbit coupling, *Phys. Rev. B* **105**, 224517 (2022).
- [25] T. Kokkeler, F. S. Bergeret, and I. V. Tokatly, Universal phenomenology of charge-spin interconversion and dynamics in diffusive systems with spin-orbit coupling, *Phys. Rev. Lett.* **134**, 096001 (2025).
- [26] T. T. Heikkilä, M. Silaev, P. Virtanen, and F. S. Bergeret, Thermal, electric and spin transport in superconductor/ferromagnetic-insulator structures, *Prog. Surf. Sci.* **94**, 100540 (2019).
- [27] F. S. Bergeret, A. F. Volkov, and K. B. Efetov, Long-Range proximity effects in superconductor-ferromagnet structures, *Phys. Rev. Lett.* **86**, 4096 (2001).
- [28] M. A. Silaev, I. V. Tokatly, and F. S. Bergeret, Anomalous current in diffusive ferromagnetic Josephson junctions, *Phys. Rev. B* **95**, 184508 (2017).
- [29] G. Schwieter, Nonlinear sigma model with particle-hole asymmetry for the disordered two-dimensional electron gas, *Phys. Rev. B* **103**, 125422 (2021).
- [30] S. Mironov and A. Buzdin, Spontaneous currents in superconducting systems with strong spin-orbit coupling, *Phys. Rev. Lett.* **118**, 077001 (2017).
- [31] F. S. Bergeret, A. F. Volkov, and K. B. Efetov, Odd triplet superconductivity and related phenomena in superconductor-ferromagnet structures, *Rev. Mod. Phys.* **77**, 1321 (2005).
- [32] “Antivortex pattern” refers here to the behavior of the charge current vector field, with a meaning similar to that used for magnetic textures. The meaning differs from antivortices in a scalar field, e.g., when discussing circulation of Abrikosov vortices, as it refers to the sign of the vector field Poincaré index, $\mathcal{P} = +1$ for vortex and $\mathcal{P} = -1$ for antivortex, instead of the sign of the winding or circulation.
- [33] V. Guillemin and A. Pollack, Oriented intersection theory, in *Differential Topology* (AMS Chelsea Publishing, Cambridge, 1974), pp. 141–149.
- [34] Vector fields and the Euler number, in *Topology from the Differentiable Viewpoint* (Princeton University Press, Princeton, 1997), pp. 33–41.
- [35] V. V. Dremov, S. Y. Grebenchuk, A. G. Shishkin, D. S. Baranov, R. A. Hovhannisyanyan, O. V. Skryabina, N. Lebedev, I. A. Golovchanskiy, V. I. Chichkov, C. Brun, T. Cren, V. M. Krasnov, A. A. Golubov, D. Rodichev, and V. S. Stolyarov, Local Josephson vortex generation and manipulation with a magnetic force microscope, *Nat. Commun.* **10**, 4009 (2019).
- [36] S. Y. Grebenchuk, R. A. Hovhannisyanyan, V. V. Dremov, A. G. Shishkin, V. I. Chichkov, A. A. Golubov, D. Rodichev, V. M. Krasnov, and V. S. Stolyarov, Observation of interacting Josephson vortex chains by magnetic force microscopy, *Phys. Rev. Res.* **2**, 023105 (2020).
- [37] J. Šoltýs, J. Feilhauer, I. Vetrova, J. Tóbiš, K. Bublikov, T. Štěpka, J. Fedor, J. Dérier, and V. Cambel, Magnetic-field imaging using vortex-core MFM tip, *Appl. Phys. Lett.* **116**, 242406 (2020).
- [38] U. Kaiser, A. Schwarz, and R. Wiesendanger, Magnetic exchange force microscopy with atomic resolution, *Nature (London)* **446**, 522 (2007).
- [39] P. Welter, J. Rhensius, A. Morales, M. S. Wörnle, C.-H. Lambert, G. Puebla-Hellmann, P. Gambardella, and C. L. Degen, Scanning nitrogen-vacancy center magnetometry in large in-plane magnetic fields, *Appl. Phys. Lett.* **120**, 074003 (2022).
- [40] D. Vasyukov, Y. Anahory, L. Embon, D. Halbertal, J. Cuppens, L. Neeman, A. Finkler, Y. Segev, Y. Myasoedov, M. L. Rappaport, M. E. Huber, and E. Zeldov, A scanning superconducting quantum interference device with single electron spin sensitivity, *Nat. Nanotechnol.* **8**, 639 (2013).
- [41] S. Jayaram, M. Lenger, D. Zhao, L. Pupim, T. Taniguchi, K. Watanabe, R. Peng, M. Scheffler, R. Stöhr, M. S. Scheurer, J. Smet, and J. Wrachtrup, Probing vortex dynamics in 2D superconductors with scanning quantum microscope, *Phys. Rev. Lett.* **135**, 126001 (2025).
- [42] S. M. Farzaneh, M. Hatefipour, W. F. Schiela, N. Lotfizadeh, P. Yu, B. H. Elfeky, W. M. Strickland, A. Matos-Abiadue, and J. Shabani, Observing magnetoanisotropic weak antilocaliza-

- tion in near-surface quantum wells, [Phys. Rev. Res. **6**, 013039 \(2024\)](#).
- [43] Z. Geng, A. Hijano, S. Ilić, M. Ilyn, I. Maasilta, A. Monfardini, M. Spies, E. Strambini, P. Virtanen, M. Calvo, C. González-Orellána, A. P. Helenius, S. Khorshidian, C. I. L. de Araujo, F. Levy-Bertrand, C. Rogero, F. Giazotto, F. S. Bergeret, and T. T. Heikkilä, Superconductor-ferromagnet hybrids for non-reciprocal electronics and detectors, [Supercond. Sci. Technol. **36**, 123001 \(2023\)](#).
- [44] H. Bai, J. Li, X. Deng, Q. Guo, X. Zhan, Y. Sun, S. Cheng, S. Xiao, P. Liu, D. Liu, J. Cai, and T. Zhu, Full-Scale field-free magnetization switching in $\text{Y}_3\text{Fe}_5\text{O}_{12}/\text{Pt}$ nanostructures for spin-orbit torque applications, [ACS Appl. Nano Mater. **8**, 50 \(2025\)](#).
- [45] T. Kontos, M. Aprili, J. Lesueur, and X. Giron, Inhomogeneous superconductivity induced in a ferromagnet by proximity effect, [Phys. Rev. Lett. **86**, 304 \(2001\)](#).
- [46] V. V. Ryazanov, V. A. Oboznov, A. Yu. Rusanov, A. V. Veretennikov, A. A. Golubov, and J. Aarts, Coupling of two superconductors through a ferromagnet: Evidence for a π junction, [Phys. Rev. Lett. **86**, 2427 \(2001\)](#).
- [47] B. Huang, G. Clark, E. Navarro-Moratalla, D. R. Klein, R. Cheng, K. L. Seyler, D. Zhong, E. Schmidgall, M. A. McGuire, D. H. Cobden, W. Yao, D. Xiao, P. Jarillo-Herrero, and X. Xu, Layer-dependent ferromagnetism in a van der Waals crystal down to the monolayer limit, [Nature \(London\) **546**, 270 \(2017\)](#).
- [48] Q. Shao, G. Yu, Y.-W. Lan, Y. Shi, M.-Y. Li, C. Zheng, X. Zhu, L.-J. Li, P. K. Amiri, and K. L. Wang, Strong Rashba-Edelstein effect-induced spin-orbit torques in monolayer transition metal dichalcogenide/ferromagnet bilayers, [Nano Lett. **16**, 7514 \(2016\)](#).
- [49] D. Crawford, S. Ilić, P. Virtanen, and T. T. Heikkilä, Numerical modelling and computer algebra analysis for Josephson anomalous vortices (2025), doi: [10.17011/jyx/dataset/107259](https://doi.org/10.17011/jyx/dataset/107259).